

Bionics Engineering, Mechanics ...

Lecture 23

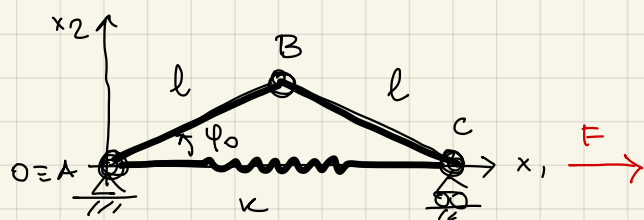
Exercises 3, 4, 4', 5

Solutions

10/xii/2025



Ex 3



$0 < \varphi_0 < \frac{\pi}{2}$ gives rest length of the spring

$$L_0 = 2l \cos \varphi_0$$

$$\underline{F}_c = F \underline{e}_1, \quad F \in (0, +\infty)$$

Kinematics: same as in Ex. 2: $\varphi = \alpha_{AB}$, $-\varphi = \alpha_{BC}$

Energy

$$L = 2l \cos \varphi, \quad L_0 = 2l \cos \varphi_0, \quad V_{el} = \frac{1}{2} k (L - L_0)^2 = 2kl^2 (\cos \varphi - \cos \varphi_0)^2$$

$$\xi_1(c) = 2l \cos \varphi - 2l \cos \varphi_0 = 2l (\cos \varphi - \cos \varphi_0), \quad W_{ext} = 2Fl (\cos \varphi - \cos \varphi_0)$$

$$V = V_{el} - W_{ext} = 2kl^2 (\cos \varphi - \cos \varphi_0)^2 - 2Fl (\cos \varphi - \cos \varphi_0)$$

Equilibrium

$$\frac{dV}{d\varphi} = 4kl^2 (\cos \varphi - \cos \varphi_0) (-\sin \varphi) - 2Fl (-\sin \varphi) = 0$$

$$(\Delta kl (\cos \varphi - \cos \varphi_0) - 2F) (\sin \varphi) = 0$$

Either $\sin \varphi = 0$, solution $\forall F \in (0, +\infty)$ $\varphi = \begin{cases} 0 \\ \pi \end{cases} + 2n\pi \Rightarrow OB \text{ and } BC \text{ are parallel to } OC \parallel \underline{e}_1$
and absorbs the difference between F and the force
in the spring $N_{AC} = k 2l (\cos \varphi - \cos \varphi_0)$

or $\sin \varphi \neq 0$ and $F = 2kl (\cos \varphi - \cos \varphi_0)$ $\left[\begin{array}{l} OB \text{ and } BC \text{ are not loaded, and} \\ F \text{ is balanced by the spring force} \\ N_{AC} = k 2l (\cos \varphi - \cos \varphi_0) \end{array} \right]$

$$\frac{F}{2kl} = \cos \varphi - \cos \varphi_0 \Leftrightarrow \frac{F}{2kl} + \cos \varphi_0 = \cos \varphi$$

Stability

$$\begin{aligned} \frac{d^2V}{d\varphi^2} &= -4kl^2 \sin \varphi (-\sin \varphi) - 4kl^2 (\cos \varphi - \cos \varphi_0) \cos \varphi + 2Fl \cos \varphi = \\ &= 4kl^2 \left(\sin^2 \varphi - \cos^2 \varphi + \cos \varphi_0 \cos \varphi + \frac{F}{2kl} \cos \varphi \right) > 0 \end{aligned}$$

• Stability / continued

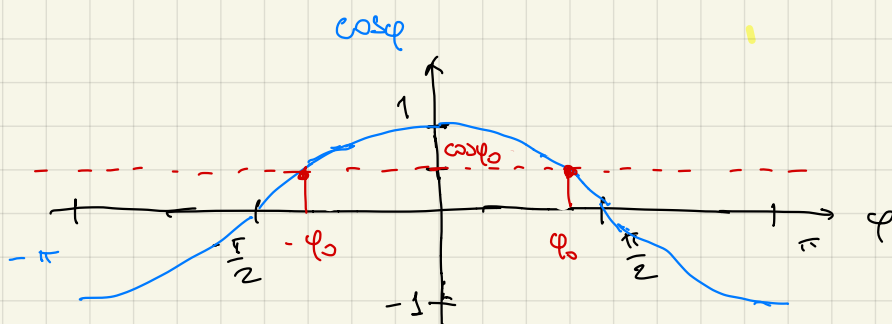
$$\begin{aligned} \frac{d^2V}{d\varphi^2} &= -4kl^2 \sin\varphi (-\sin\varphi) - 4kl^2 (\cos\varphi - \cos\varphi_0) \cos\varphi + 2Fl \cos\varphi = \\ &= 4kl^2 \left(\sin^2\varphi - \cos^2\varphi + \cos\varphi_0 \cos\varphi + \frac{F}{2kl} \cos\varphi \right) > 0 \end{aligned}$$

$$\bullet \left. \frac{d^2V}{d\varphi^2} \right|_{\varphi=0} > 0 \Leftrightarrow -1 + \cos\varphi_0 + \frac{F}{2kl} > 0 \Leftrightarrow \frac{F}{2kl} > 1 - \cos\varphi_0$$

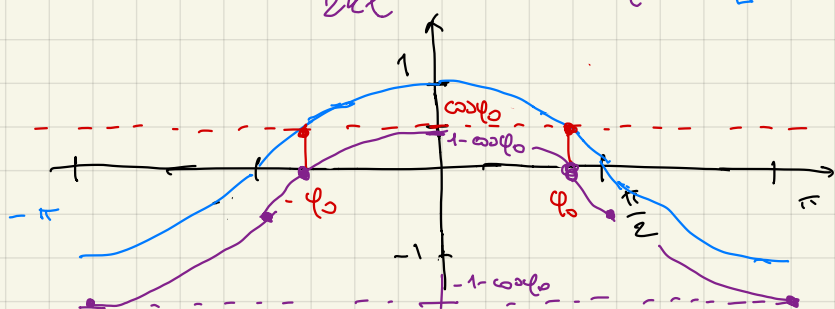
$$\begin{aligned} \bullet \frac{F}{2kl} = \cos\varphi_{eq} - \cos\varphi_0 &\Rightarrow \left. \frac{d^2V}{d\varphi^2} \right|_{\varphi=\varphi_{eq}} = 4kl^2 \left(\sin^2\varphi_{eq} - \cancel{\cos^2\varphi_{eq}} + \cancel{\cos\varphi_0} \cancel{\cos\varphi_{eq}} + \right. \\ &\quad \left. + \cancel{\cos\varphi_{eq}}^2 - \cancel{\cos\varphi_{eq}} \cancel{\cos\varphi_0} \right) \\ &= 4kl^2 \sin^2\varphi_{eq} > 0 \quad \text{always true} \end{aligned}$$

• Interpretation and F vs φ diagram (a gentle reminder)

$$\frac{F}{2kl} = \cos\varphi - \cos\varphi_0, \quad 0 < \varphi_0 < \frac{\pi}{2}$$



$$\frac{F}{2kl} = \cos\varphi - \cos\varphi_0 \quad [\text{translate } \cos\varphi \text{ downwards by } \cos\varphi_0]$$



• $\varphi = 0$

$$\frac{F}{2kl} = (\cos\varphi - \cos\varphi_0) \Big|_{\varphi=0} = 1 - \cos\varphi_0$$

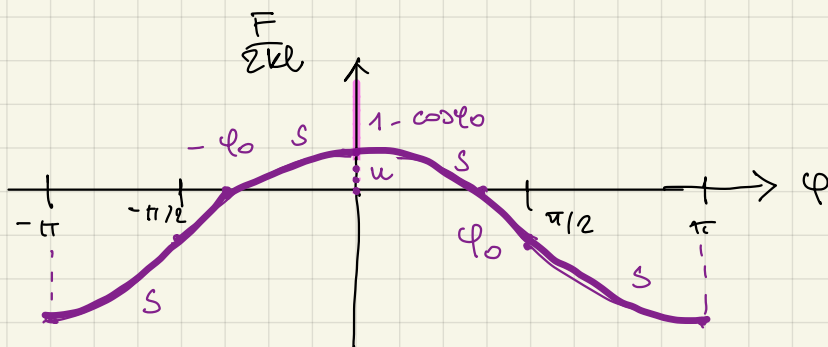
$$\Rightarrow F = 2kl(1 - \cos\varphi_0) > 0$$

• $\varphi \neq \varphi_0$

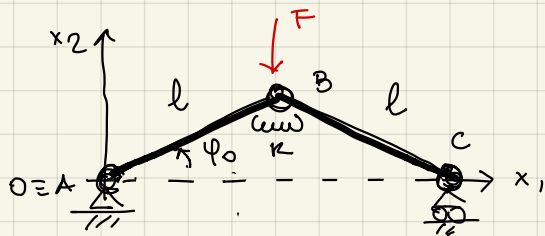
$$\frac{F}{2kl} = (\cos\varphi - \cos\varphi_0) \Big|_{\varphi \neq \varphi_0} = 0 \Rightarrow F = 0$$

• $|\varphi| > \varphi_0 \Rightarrow F < 0$

• Force vs φ diagram (Recall: $\varphi = 0$ is equilibrium $\forall F \in (0, +\infty)$)



Ex 4



$$\underline{F}_B = -F \underline{e}_2, \quad F \in (0, +\infty)$$

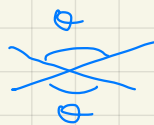
$$V_{el} = \frac{1}{2} k (\Delta\varphi - \Delta\varphi_0)^2$$

Kinematics

Same as Ex 2, 3 : $\alpha_{AB} = \varphi$, $\alpha_{BC} = -\varphi$

$$\left. \begin{aligned} \Delta\varphi_0 &= -\varphi_0 - \varphi_0 = -2\varphi_0 \\ \Delta\varphi &= -\varphi - \varphi = -2\varphi \end{aligned} \right\} \Rightarrow \Delta\varphi - \Delta\varphi_0 = (-2\varphi + 2\varphi_0)$$

Alternatively: internal angles



$$\left. \begin{aligned} \Theta_0 &= \pi - 2\varphi_0 \\ \Theta &= \pi - 2\varphi \end{aligned} \right\} \Rightarrow \Theta - \Theta_0 = (\pi - 2\varphi - \pi + 2\varphi_0)$$

$$B(\varphi_0) = (l \cos \varphi_0, l \sin \varphi_0)$$

$$B(\varphi) = (l \cos \varphi, l \sin \varphi)$$

Energy

$$V_{el} = \frac{1}{2} k (\Theta - \Theta_0)^2 = \frac{1}{2} k (\Delta\varphi - \Delta\varphi_0)^2 = 2k (-\varphi + \varphi_0)^2$$

$$W_{ext} = (0, -F) \cdot (l(\cos \varphi - \cos \varphi_0), l(\sin \varphi - \sin \varphi_0)) = -Fl(\sin \varphi - \sin \varphi_0)$$

$$V = V_{el} - W_{ext} = 2k (-\varphi + \varphi_0)^2 - Fl(\sin \varphi_0 - \sin \varphi)$$

Equilibrium

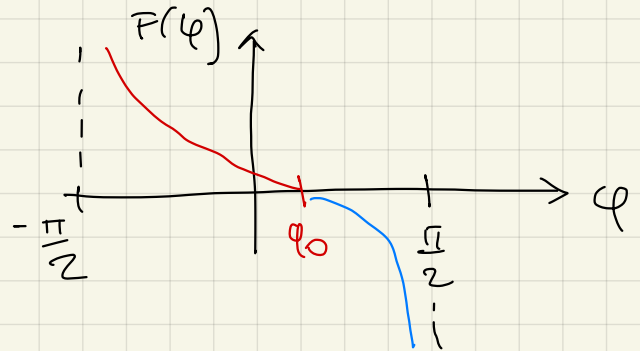
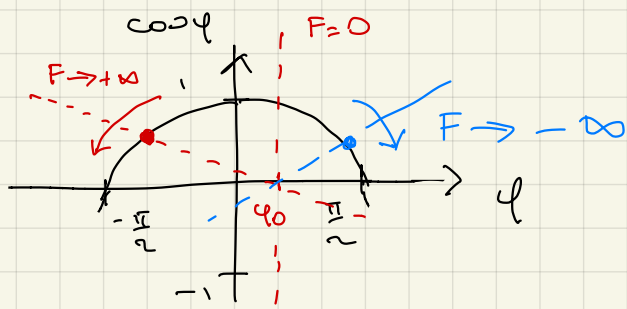
$$\frac{dV}{d\varphi} = -4k(-\varphi + \varphi_0) + Fl \cos \varphi = 0$$

Either $F=0$, then $-4k(-\varphi + \varphi_0) = 0 \Rightarrow \varphi = \varphi_0$, only solution

$$\text{or } F \neq 0, \text{ then } \cos \varphi = \frac{4k}{Fl} (-\varphi + \varphi_0)$$

Either $F=0$, then $-4k(-\varphi+\varphi_0)=0 \Rightarrow \varphi = \varphi_0$, only solution

or $F \neq 0$, then $\cos \varphi = -\frac{4k}{F\ell} (\varphi - \varphi_0)$



$\forall F \in (0, +\infty)$ there is only one equilibrium in the range $\varphi \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$\varphi_{eq} \rightarrow -\frac{\pi}{2}$ when $F \rightarrow +\infty$

• stability

$$\frac{d^2V}{d\varphi^2} = 4k - F\ell \sin \varphi > 0$$

$$\Leftrightarrow -\frac{4k}{F\ell} < -\sin \varphi = \frac{d}{d\varphi} (\cos \varphi) \text{ always true.}$$

Indeed,

• For $\varphi_{eq} \in (0, \varphi_0)$: $\sin \varphi > 0$ but $-\frac{4k}{F\ell} < -\sin \varphi < 0$

by inspection of the slopes in the diagram $\cos \varphi$ vs φ

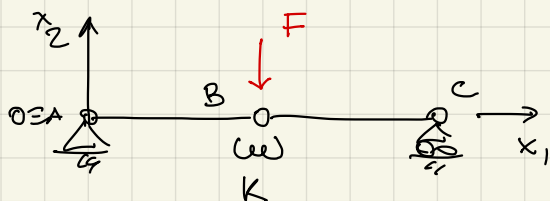
• For $\varphi_{eq} \in (-\frac{\pi}{2}, 0)$: $\sin \varphi < 0$ and $-\frac{4k}{F\ell} < 0 < -\sin \varphi$

Remark

The case $F \in (-\infty, 0)$ is similar, look at blue graph of $-\frac{4k}{F\ell} (\varphi - \varphi_0)$

$\varphi \rightarrow \frac{\pi}{2}$ when $F \rightarrow -\infty$

Ex 4'

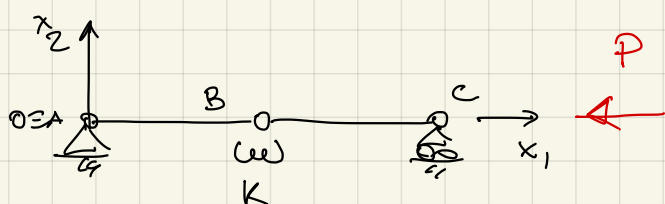


$$\underline{F}_B = -F \underline{e}_2$$

$$V_{el} = \frac{1}{2} k (\Delta \varphi)^2$$

Just set $\varphi_0 = 0$ in Ex. 4

Ex 5



$$\underline{F}_C = -P \underline{e}_1, \quad P \in (-\infty, +\infty)$$

$$V_{el} = \frac{1}{2} k (\Delta \varphi)^2$$

Restrict attention to $\varphi \in (-\pi, \pi)$

. Kinematics

Same as Ex 2, 3 : $\alpha_{AB} = \varphi, \quad \alpha_{BC} = -\varphi$

$$\Delta \varphi_0 = 0$$

$$\Delta \varphi = -\varphi - \varphi = -2\varphi \quad \left. \vphantom{\Delta \varphi} \right\} \Rightarrow \Delta \varphi^2 = 4\varphi^2$$

. Energy

$$V_{el} = \frac{1}{2} k (\Delta \varphi)^2 = 2k\varphi^2$$

$$W_{ext} = P(2l - 2l \cos \varphi) = 2Pl(1 - \cos \varphi)$$

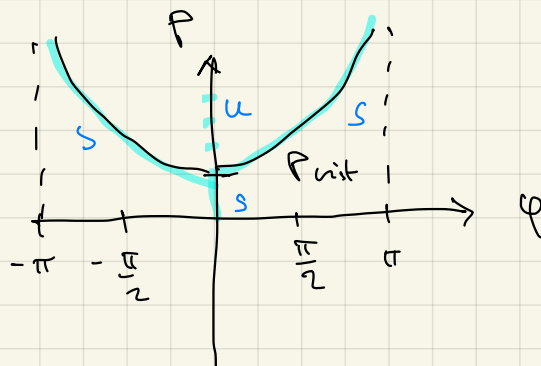
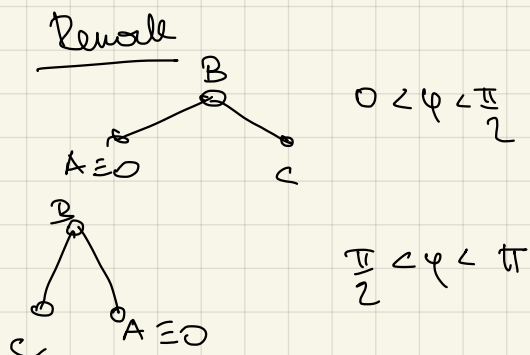
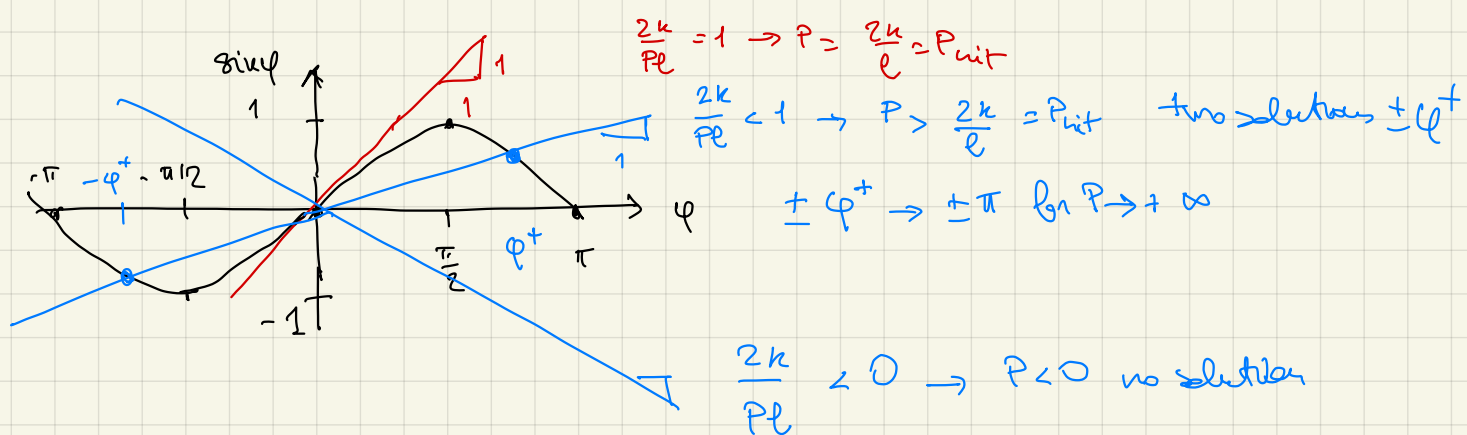
$$V = V_{el} - W_{ext} = 2k\varphi^2 - 2Pl(1 - \cos \varphi) = 2k\varphi^2 + 2Pl(\cos \varphi - 1)$$

. Equilibrium

$$\frac{dV}{d\varphi} = 0 = 4k\varphi - 2Pl \sin \varphi = 0$$

Either $\sin \varphi = 0 \rightarrow \varphi = 0$ solution $\forall P \in (-\infty, +\infty)$

or $\sin \varphi \neq 0 \rightarrow P = \frac{2k}{l} \frac{\varphi}{\sin \varphi} \neq 0$ and $\frac{2k}{Pl} \varphi = \sin \varphi$



• Stability

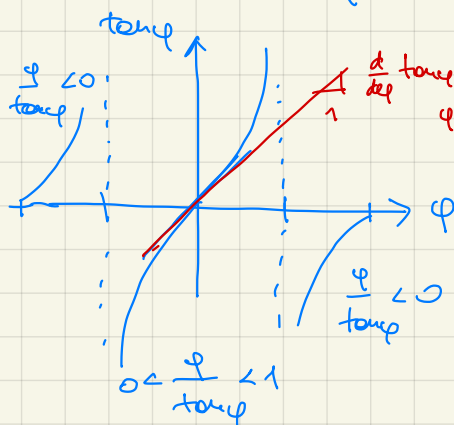
$$\frac{d^2V}{d\varphi^2} = 4k - 2Pl \cos \varphi > 0$$

For $\varphi = 0$ $\frac{d^2V}{d\varphi^2} \Big|_{\varphi=0} = 4k - 2Pl > 0 \Leftrightarrow P < \frac{2k}{l} = P_{crit}$

For $P > 0$, $\frac{d^2V}{d\varphi^2} > 0 \Leftrightarrow \cos \varphi = \frac{d}{d\varphi}(\sin \varphi) < \frac{2k}{Pl}$ always true by inspection of the graph of $\sin \varphi$

Alternatively: plug $P_{eq} = \frac{2k}{l} \frac{\varphi}{\sin \varphi}$ in $\frac{d^2V}{d\varphi^2}$:

$$4k - 2 \frac{2k}{l} \frac{\varphi}{\sin \varphi} l \cos \varphi = 4k - 4k \frac{\varphi}{\tan \varphi} = 4k \left(1 - \frac{\varphi}{\tan \varphi}\right) > 0$$



$\forall \varphi \in (-\pi, 0) \cup (0, \pi)$

Check:

- $\varphi \in (-\frac{\pi}{2}, \frac{\pi}{2}) \setminus 0$, see graph

- $\varphi \in (-\pi, -\frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi)$ because $\frac{\varphi}{\tan \varphi} < 0$